CPSC 375
Introduction to Data Science and Big Data Analytics

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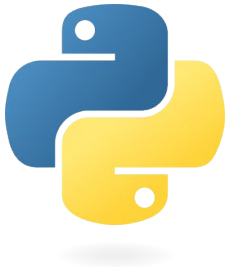
Week 01 L02

Statistical software

- SAS
- SPSS
- Matlab
- S-Plus
- R
- …

History of Python

- Object-oriented scripting language
  - extensible with C or C++
  - developed at Dutch National Research Institute
  - **"There should be one– and preferably only one– obvious way to do it."** – Guido van Rossum, creator of Python



- Python in Data Science:
  - scientific computing
  - in 2006, first release of NumPy library
  - huge ecosystem: pandas, scikit-learn, statsmodels, …

Python: Main features

- General purpose programming language
  Object-oriented
- NumPy library implements vector-based operations
  - Similar to MATLAB

•Packaging: easily add software libraries from different sources

•And features of typical programming languages

Python and statistics

- Many packages for statistics and data analysis
  - linear models and generalized linear models
  - nonlinear regression models
  - time series analysis
  - classical parametric and nonparametric tests
  - clustering
  - smoothing
- And newer statistical methods
  - many data scientists provide their methods as Python packages

Python and Presentations

- Many functions for creating various types of data presentations
- Streamlit, Dash, Shiny
  - Python packages to build interactive web apps for data science
  - Host standalone apps on a webpage
  - Packages for general-purpose web development (Django, Flask, FastAPI)
- Jupyter Notebooks
  - Authoring framework for data science
  - A single Jupyter notebook file to both
    - save and execute code
    - generate reports
      - HTML
      - PDF
      - …

Strengths

- Free and Open Source
- Strong user community
- Maintained by experts
- Highly extensible
- Continuous improvement
- **Core language is easy to learn**
- Available on many platforms

Weaknesses

- Slow for large datasets
- All data has to fit in memory (RAM)
  - Data limited to RAM size (e.g., 8GB)
  - Big data platforms also support Python
- Steep learning curve for libraries
  - NumPy, SciPy, Pandas, scikit-learn, statsmodels, matplotlib, plotly, letsplot, Tensorflow, PyTorch, …
- Data type conversion between libraries
  - Python lists, NumPy arrays, Pandas DataFrames, …

Installation – Local Machine

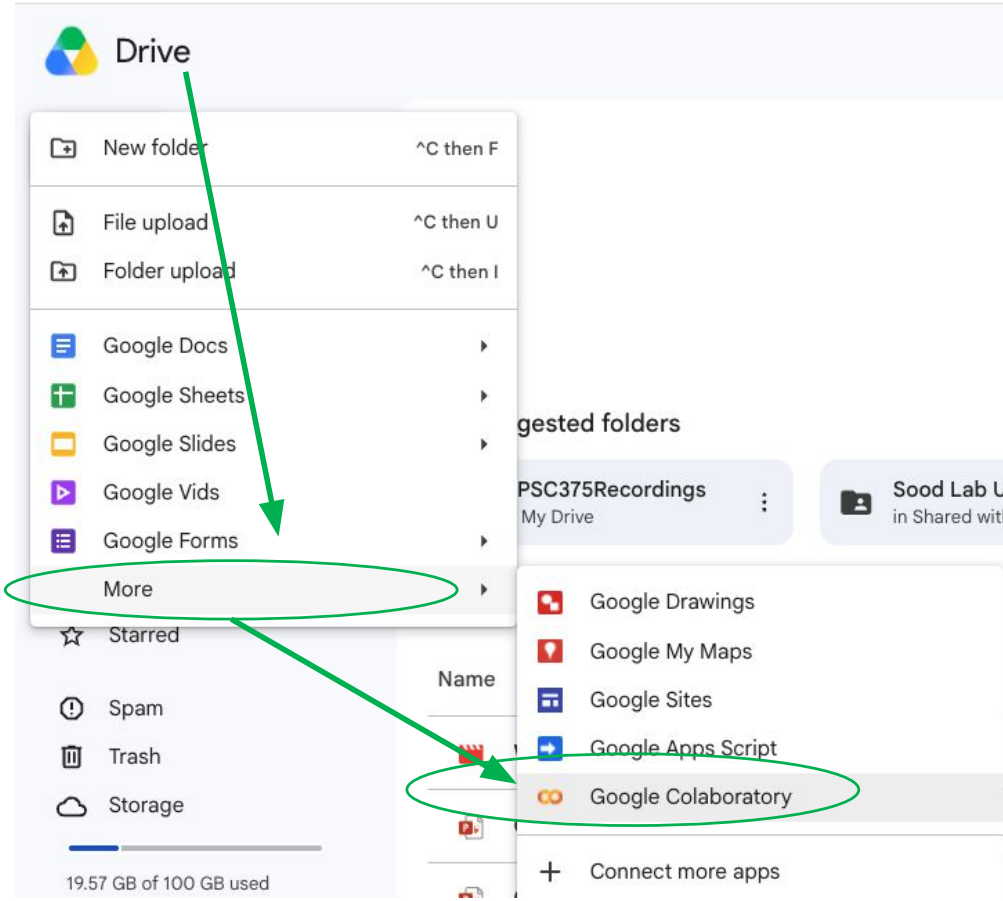
- Three steps

1. Install the Anaconda Distribution
   - Interpreter, libraries, package manager
   - Binaries for Windows/Linux/MacOS

2. Install Visual Studio Code
3. Configure Visual Studio Code Extensions
   - See *Python for Data Science*, Prerequisites for details.

Installation – Cloud

- Required for classwork
- Two steps
  1. Log into Google Drive
     - for classwork, use your @csu.fullerton.edu account
     - You may need to Switch between multiple Google accounts
  2. Choose *New* → *More* → *Google Colaboratory*



Getting help

Details about a specific module whose name you know (high-level overview, list of functions):

```
>>> import scipy.stats
>>> help(scipy.stats)
```

Details about a specific command whose name you know (input arguments, options, algorithm, results):

```
>>> help('scipy.stats.ttest_1samp')
Help on function ttest_1samp in scipy.stats:

scipy.stats.ttest_1samp = ttest_1samp(a, popmean, axis=0, nan_policy='propagate',
alternative='two-sided', *, keepdims=False)
    Calculate the T-test for the mean of ONE group of scores.
...
```

From a code cell in a Jupyter notebook:

```
import scipy.stats
?scipy.stats
```

Arithmetic operations

- + (addition)
- - (subtraction)
- * (multiplication)
- / (division)
- // (integer division)
- ** (exponentiation)

Basic functions

- `abs(x)`

`from math import...`

- `log(x)`
- `log2(x)`
- `log10(x)`
- `exp(x)`
- `sqrt(x)`

Assignment

```
 a = 20 # assign 20 to object "a"
 a # print the value of object
20
```

Data types

- int
  - 124
- float
  - 3.141
- str
  - "strings" or 'strings'
- bool
  - True, False

Data Structures in Python

	1-dim	2-dim
All same type	**ndarray (NumPy)**	**matrix (NumPy)**
Mixed types	**list**	**DataFrame (Pandas)**

Vectors

- ndarray: an ordered collection of data of the same type

```
import numpy as np
a = np.array([10,20,30])
```

- Subscripts in square brackets
  - a[0] is the first element in vector a

Vectorized functions

- Most NumPy functions take an “array_like” object as input
- Answer can be a vector or numeric
  - `np.sum(np.array([1, 2, 3]))`
- Most operators are vectorized too
  - `a*2`
  - `array([2, 4, 6])`
  - `np.array([True, False, True]) & np.array([True,True,False])`
  - `array([ True, False, False])`

Sequences and Slices

- np.arange(1, 4)
  - array([1, 2, 3])
- np.arange (1,4,2) == np.arange(1,4,step=2)
- np.linspace (1,20,num=8)

- Can be used for numeric indexing
  - a[np.arange(2,6)] is elements at index 2-5 in vector a
  - “Slice” syntax a[2:6] as shortcut

- Caution:
  - Arrays begin at index 0
  - Last element is *stop* - 1

Opening classwork

- https://github.com/soodk6/cpsc375-classwork
  - python.ipynb – Part 1

Links on Canvas

- From GitHub
  - View notebook
  - Click *Open in Colab*
- From Google Colaboratory
  - Open notebook ⟶ GitHub
  - User *soodk6*, repository *soodk6/cpsc375-classwork*
  - View notebook

Image Courtesy: Gemini

Submitting classwork

- Check that you are logged in with your csu.fullerton.edu email address
  - Click your profile photo or email address at the top right corner
    - Check that it shows your @csu.fullerton.edu email address at the top
  - If not, choose that account or click *Add Account*
- Click *Copy to Drive* or *File → Save a Copy in Drive*
- Rename the notebook with your groupmates' lastnames
  - E.g.: Doe-Chester-Hoffman.ipynb
  - Write the names and role of group members
  - Roles: manager, recorder+presenter, timekeeper
  - Add your answers
- Click *File → Move*
  - Change folder to *Shared with Me* → CPSC375StudentFolder-Term → today's date
- Will not be graded may be used for class participation points
  - Can easily share with rest of class

Classwork - Suggested Workflow

- Multiple users can open the same notebook, but content sync is not as seamless as Google Docs
  - you may receive a notification that the notebook has changed, but you will need to reload the page.
- Suggestion for classwork:
  - *Recorder* opens notebook from GitHub and saves answers to Google Drive
  - Other group members open notebook from GitHub but do not save to Google Drive.
    - Experiment with a temporary notebook
    - Colab will say “Cannot save changes” until you make a copy.

Lists and Dictionaries

- list: an ordered collection of data of arbitrary types.

```
>>> course = ["CPSC", 375, False]
>>> course[0]
'CPSC'
```

- dict: an ordered collection of key-value pairs.

```
>>> course = dict(dept="CPSC", number=375, required=False)
```

or

```
>>> course = {'dept':"CPSC", 'number':375, 'required':False}
>>> course['dept']
'CPSC'
>>> course["number"]
375
```

- Typically
  - list elements are accessed by their index (an integer)
  - dict elements by their key (usually a character string)

Matrices and Arrays

- numpy.matrix: a rectangular table of data of the same type
  - 2-dimensional
  - * for matrix multiply
- numpy.ndarray: *n*-dimensional matrix ($n \geq 1$)
  - * is element-wise
  - @ for matrix multiply

1	2	31	2	9	7	34	22	11	5
11	92	4	3	2	2	3	3	2	1
3	9	13	8	21	17	4	2	1	4
8	32	1	2	34	18	7	78	10	7
9	22	3	9	8	71	12	22	17	3
13	21	21	9	2	47	1	81	21	9
21	12	53	12	91	24	81	8	91	2
61	8	33	82	19	87	16	3	1	55
54	4	78	24	18	11	4	2	99	5
13	22	32	42	9	15	9	22	1	21

Data frames (pandas.core.frame.DataFrame)

- Typical data table from a spreadsheet
- Table with rows and columns
- Each column has the same type (e.g. number, text, logical),
- Different columns may have different types

- Example:

```
   localisation   tumorsize     progress
0     proximal         6.3      False
1       distal         8.0       True
2     proximal        10.0      False
```

Create the above DataFrame?

```
import pandas as pd
pd.DataFrame({'localisation': ['proximal', 'distal', 'proximal'],
'tumorsize': [6.3, 8.0, 10.0], …)
```

Accessing parts of the data

- Rows are also called observations or cases
- Columns are also called variables or series
- The data table is an object
- Subscripts with iloc ("integer location") in square brackets
  - y.iloc[0] identifies first row in matrix/data.frame y
  - y.iloc[0:5] identifies first 5 rows
  - y.iloc[:,0] identifies first column
  - y.iloc[2,1] element in row 2 and column 1
  - y.iloc[1:5,2:4] elements in rows 1-5 and columns 2-4

Subsetting

Note: row numbers (not a column) are not included in size

	carat	cut	color	clarity	depth	table	price	x	y	z
1	0.52	Ideal	D	VS2	61.4	56	1664	5.16	5.19	3.18
2	0.5	Very Good	F	SI1	62.3	60	1250	5.07	5.11	3.17
3	0.61	Ideal	G	VVS2	61.6	54	2242	5.45	5.49	3.37
4	0.36	Premium	G	VS2	62.5	58	756	4.55	4.51	2.83
5	0.7	Very Good	E	VS2	63.5	54	2889	5.62	5.66	3.58
6	0.56	Ideal	F	VS1	61.7	56	2016	5.32	5.28	3.27
7	1.19	Premium	E	I1	60.2	61	3572	6.91	6.87	4.15
8	0.52	Ideal	F	IF	60.6	57	2575	5.21	5.22	3.16
9	0.38	Ideal	E	IF	62.7	55	1433	4.61	4.67	2.91
10	0.51	Ideal	E	VS2	62.1	54	1608	5.13	5.15	3.19

Note: column/variable names are not included in size

df.loc[3] row 4

df.iloc[5,6] element in row 6 and column 7

df.iloc[:, 3]
df.clarity
df["clarity"]
column 4

df.iloc[7:10, 7:9]
df.**loc**[7:10,("x", "y")]

Logical Indexing

- d[d<20]: extract all elements of d that are smaller than 20
- d<20
  - A logical vector (True/False)
- Convert from logical to numeric indexes:
  - `np.where()`
  - `x = np.arange(1,26,step=3)`
  - `x==7: condition`
  - `np.where(x==7)`

Categoricals

- A character string can contain arbitrary text.
- A categorical is a variable that can only take a limited number of values, which are called categories.
- Limit the vocabulary, with a small number of allowed words

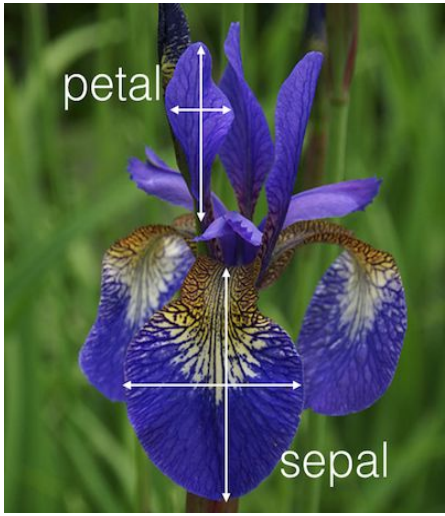
```
>>> iris.Species.dtype
CategoricalDtype(categories=['setosa', 'versicolor',
'virginica'], ordered=False, categories_dtype=object)
```

The Iris Flower Dataset

Iris setosa

Iris versicolor

Iris virginica



Sepal /Petal

https://en.wikipedia.org/wiki/Iris_flower_data_set
http://sebastianraschka.com/Articles/2014_python_lda.html

The iris dataset

- Show first 10 rows
- Show column Sepal.Length
- What is the mean Sepal.Length?
- Show all rows where Sepal.Length > 7.6
  - What are the row indexes where Sepal.Length > 7.6?

Classwork

- soodk6/cpsc375-classwork
  – python.ipynb – Part 2

Coercion

When different objects are mixed in a vector, coercion occurs so that every element in the vector is of the same class:

```
>>> np.array([1,"a"]) ## character
array(['1', 'a'], dtype='<U11')
>>> np.array([True,2]) ## numeric
array([1, 2])
>>> np.array(["a",True]) ## character
array(['a', 'True'], dtype='<U5')
```

Explicit coercion

Objects can be explicitly coerced from one class to another using the astype method:

```
>>> x=np.arange(1,6)
>>> x.dtype
dtype('int32')

>>> x.astype(float)
array([1., 2., 3., 4., 5.])

>>> x.astype(bool)
array([ True,  True,  True,  True,  True])

>>> x.astype(str)
array(['1', '2', '3', '4', '5'], dtype='<U11')
```

Some special values

- `np.NaN`: Not a Number (IEEE 754, used by Pandas as placeholder for a missing value).
- `pd.NaT`, `pd.NA` : Not a Time, Not Available (placeholders for missing values).
- `None`: empty value (Python equivalent of C NULL, C++ nullptr)
- `np.Inf`: Infinity (IEEE 754)
- `pd.isna()` and `np.isinf()`
  - to identify missing and infinite values in data
  - Pandas treats None as missing
- `df.notna().all(axis='columns')` to identify the rows in a data frame that are complete (no missing values)
- Many functions give NaN if *any* of the inputs is NaN, e.g.: `np.mean(x)`
  - Some functions have nan* versions to ignore NaNs, e.g.: `np.nanmean(x)`
  - Pandas methods ignore NA values by default, e.g.: `iris['Sepal.Length'].mean()`
  - Give parameter `skipna=False` to retain the NAs in the input

Classwork

- soodk6/cpsc375-classwork
  –python.ipynb – Part 3

Storing (in memory) variables

- Every Pandas object can be stored into a file with the method “to_pickle” and restored from a file with the function “read_pickle”.
- This uses a Python-specific format that is portable between MS-Windows, Linux, Mac.

```
>>> x.to_pickle("x.pkl")
>>> pd.read_pickle ("x.pkl")
```

Questions?